침구의학 SUMMARY 공지사항

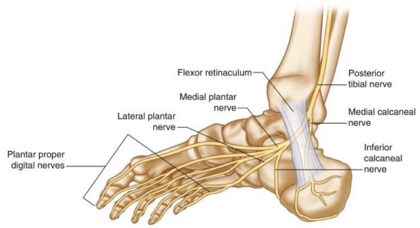
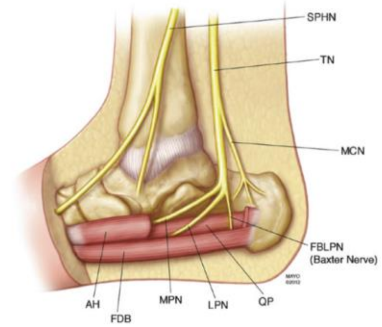
- 본시험: 말초신경 10개 선정하여 거기세어 60-80% 출제, 객관식 30문제로 CBT 형식으로 출제(증상을 묻고 포착된 말초신경을 묻거나 치료부위(혈자리)를 묻는 방식
- 예비 시험: 5/31 금요일 2교시(10시~), CBT실, 수업 한시간 하고 2교시에 함께 이동하여 시행, 성적 반영 안하고 응시해야 출석으로 간주, 5문항 정도 나오며, 기말고사에 꼭 출제할 예정
- 혈자리를 시험에 내시겠다고 하셨는데.. 수업 시간에 혈 언급이 별로 없어서 아무래도 경혈들의 유주 분포를 암기해야 될 것으로 보입니다.

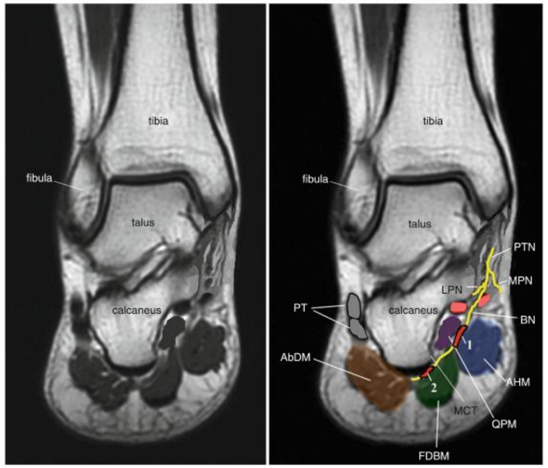
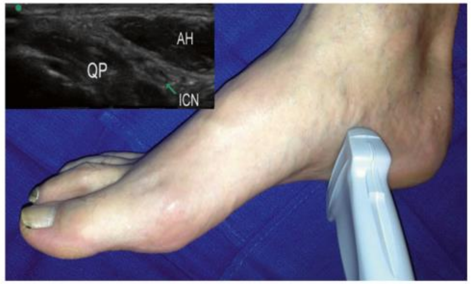
침구의학 SUMMARY 사용법

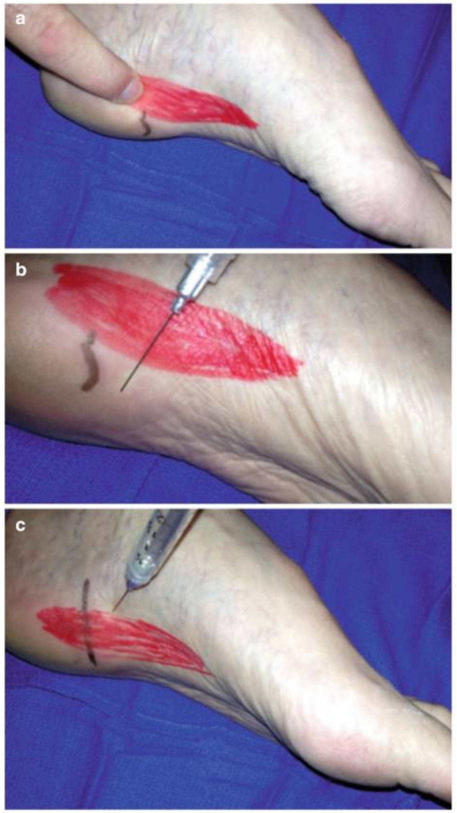
- 새로운 교수님으로 이전 학번 기출이 없습니다.
- 수업 내용이 깔끔하고, 어렵지 않기 때문에 공부해보시는 것을 추천드립니다.
- 중요해 보이는 내용은 가독성을 위해 **굵은 글씨**와 빨간 글씨로 표시하겠습니다.
- 사진이 너무나 많아서 적당한 분량을 위해 사이즈를 줄였습니다.

6절. 하지부 포착신경 병증

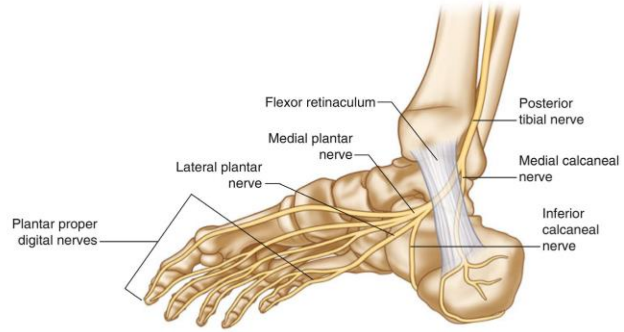
16. 종골하신경 포착신경병증 (inferior calcaneal nerve entrapment)

증상	•내측 발뒤꿈치 통증 •족궁이나 발목으로 방사되는 통증 •오래 걷거나 일어서 있을 때 악화 •Baxter's nerve라고도 함 
원인	•오래 걷거나 서있는 직업 •외상이나 수술 •발의 과도한 회내
포착 위치	•Between the dense deep fascia of the AbHand the medial aspect of the QP
	•Where the ICN crosses anterior to the MCT (or heel spur if one is present) - Baxter's nerve 확인할 수 있다.  **Fig. 76.5** Drawing of the medial ankle showing the branching pattern of the tibial nerve in the proximal and distal tarsal tunnels. The abductor hallucis (AbH) has been partially removed to show the vertical course of the inferior calcaneal nerve (ICN) and the interval between the AbH (removed) and quadratus plantae muscles (QP). *SPHN* saphenous nerve, *TN* tibial nerve, *MCN* medial calcaneal nerve, *FBLPN* first branch of the lateral calcaneal nerve (inferior calcaneal nerve), *QP* quadratus plantae, *LPN* lateral plantar nerve, *MPN* medial plantar nerve, *FDB* flexor digitorum brevis, *AH* abductor hallucis. Note the vertical course of the ICN perpendicular to the parallel paths of the AH, FDB, and QP (From Presley et al. [5]. Reprinted with permission from American Institute of Ultrasound in Medicine)
통증 부위	- 발안쪽, 발뒤꿈치 쪽을 눌렀을 때 통증이 발생한다. - **족저근막염과 해당 신경 포착을 감별**해야 하는데,  **Fig. 76.1** Patient pain complaint from inferior calcaneal nerve entrapment (Image courtesy of Andrea Trescot, MD)  **Fig. 76.2** Tenderness of the medial calcaneal tubercle (Image courtesy of Andrea Trescot, MD) 족저근막염은 아침에 일어나서 걸을 때, 활동할 때 족저근막이 늘어나 통증이 감소한다. 반대로 신경 포착의 경우는 압박이 심할 때, 활동이 많아지면 심해진다.

영상	 **Fig. 76.6** Locations of inferior calcaneal nerve entrapment. *PTN* posterior tibial nerve, *MPN* medial plantar nerve, *LPN* lateral plantar nerve, *BN* Baxter's nerve (inferior calcaneal nerve), *AHM* abductor hallucis muscle, *QPM* quadratus plantae muscle, *FDBM* flexor digitorum brevis muscle, *AbDM* abductor digiti minimi muscle, *MCT* medial calcaneal tubercle, *PT* peroneus (fibularis) brevis and longus tendons. *1* Entrapment site between the AHM and QPM. *2* Entrapment site between the MCT and the FDBM (Image courtesy of Andrea Trescot, MD) - 노란색으로 신경이 주행하는 것을 확인할 수 있다. **Fig. 76.10** In-plane ultrasound image showing the probe location and the neurovascular bundle that includes the inferior calcaneal nerve. *AH* adductor hallucis muscle, *QP* quadratus plantae muscle, *ICN* inferior calcaneal nerve (Image courtesy of Michael Brown, MD)   **Fig. 76.14** In-plane ultrasound technique of ICN injection (Image courtesy of Michael N. Brown, MD)  **Fig. 76.13** Composite images of the heel structures, coronal cross-sectional technique. (**a**) MRI cross section with simulated needle; (**b**) ultrasound transverse image; *ADM* abductor digiti minimi muscle, *AH* abductor hallucis, *FHB* flexor hallucis brevis, *QP* quadratus plantae (Image courtesy of Brian Shilpe, DO)

추가 내용	 **Fig. 76.8** Landmark-guided inferior calcaneal nerve injection. (**a**) palpation of injection site; (**b**) depth of injection; (**c**) inferior calcaneal nerve injection; *Red* structure, outline of the abductor hallucis muscle (Image courtesy of Michael N. Brown, MD) - 사진과 같이 통증이 유발되는 부위를 잡고 치료한다. - 발바닥이라 엄청 아프기에 길이와 굵기가 적당한 주사를 사용한다.

17. 내측 종골신경 포착신경병증 (medial calcaneal nerve entrapment)

<table>
<tr><td>증상</td><td>•발뒤꿈치의 날카로운 신경통
•오래 서있거나 걸을때 통증악화
•신경인성 발뒤꿈치 통증의 흔한 원인


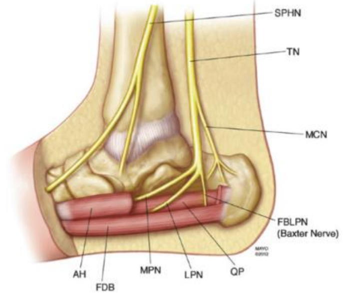
- 발목에서 가장 많이 포착되는 신경</td></tr>
<tr><td>원인</td><td>•발뒤꿈치의 날카로운 신경통
•오래 서있거나 걸을때 통증악화
•신경인성 발뒤꿈치 통증의 흔한 원인


Fig. 76.5 Drawing of the medial ankle showing the branching pattern of the tibial nerve in the proximal and distal tarsal tunnels. The abductor hallucis (AbH) has been partially removed to show the vertical course of the inferior calcaneal nerve (ICN) and the interval between the AbH (removed) and quadratus plantae muscles (QP). *SPHN* saphenous nerve, *TN* tibial nerve, *MCN* medial calcaneal nerve, *FBLPN* first branch of the lateral calcaneal nerve (inferior calcaneal nerve), *QP* quadratus plantae, *LPN* lateral plantar nerve, *MPN* medial plantar nerve, *FDB* flexor digitorum brevis, *AH* abductor hallucis. Note the vertical course of the ICN perpendicular to the parallel paths of the AH, FDB, and QP (From Presley et al. [5]. Reprinted with permission from American Institute of Ultrasound in Medicine)</td></tr>
</table>

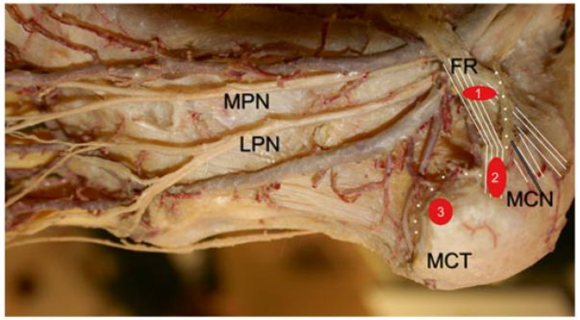
<table>
<tr><td>포착 위치</td><td>•Proximal tarsal tunnel
•AbHhypertrophy
•Medial corner of the MCT
- 모두 어떤 방식으로든 압박을 받을 수 있는 구조물들이다.


Fig. 77.5 Dissection of the medial ankle, from *Bodies, The Exhibition*, with permission. Circles represent possible sites of entrapment. Note the medial calcaneal nerve coming off *proximal to* the flexor retinaculum and the cutaneous plantar nerve (anterior terminal branch) following the contour of the calcaneus to the medial aspect of the medial calcaneal tubercle. Note also the position of the posterior tibial nerve and its branches deep to the blood vessels. The abductor hallucis muscle has been removed. *FR* flexor retinaculum, *MCN* medial calcaneal nerve (*white dots*), *MCT* medial aspect of the medial calcaneal tubercle, *LPN* lateral plantar nerve, *MPN* medial plantar nerve (Image courtesy of Andrea Trescot, MD)</td></tr>
<tr><td>통증 부위</td><td>- 신경주행이 발바닥에서 발뒤꿈치까지 이어지기에 해당 부위들에서 통증이 유발된다.

Fig. 77.1 Patient pain complaint of heel pain from medial calcaneal nerve entrapment (Image courtesy of Andrea Trescot, MD)

Fig. 77.2 Pattern of pain from medial calcaneal nerve entrapment (Image courtesy of Andrea Trescot, MD)
the flexor retinaculum) at the medial ankle where it divides</td></tr>
</table>

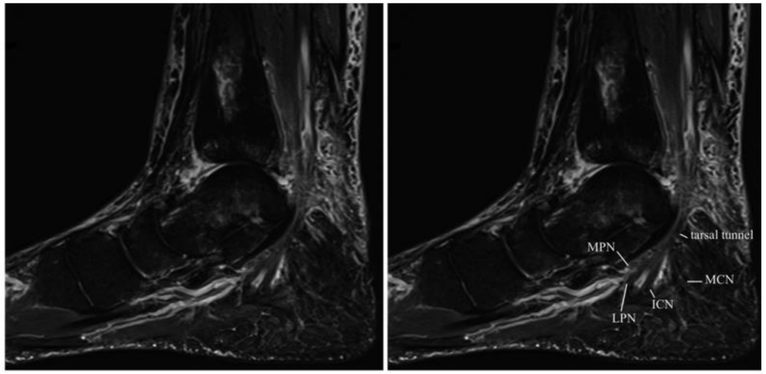
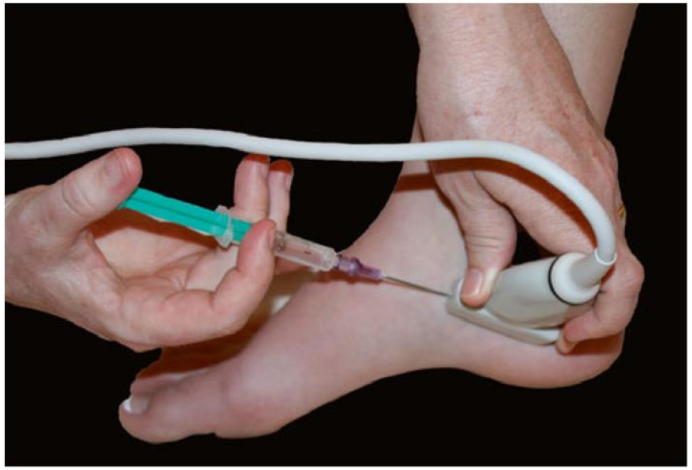
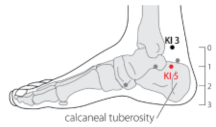
영상	 **Fig. 77.11** STIR sagittal MRI of the medial malleolus showing the tarsal tunnel. *MCN* medial calcaneal nerve, *ICN* inferior calcaneal nerve, *LPN* lateral plantar nerve, *MPN* medial plantar nerve (Image courtesy of Andrea Trescot, MD)
추가 내용	 **Fig. 77.15** Ultrasound for in-plane injection of the medial calcaneal nerve (Image courtesy of Andrea Trescot, MD)

	 **Fig. 77.9** Palpation of the medial calcaneal nerve at the proximal abductor hallucis muscle (Image courtesy of Michael Brown, MD)  **Fig. 77.12** Landmark-guided medial calcaneal nerve injection (Image courtesy of Gabor Racz, MD)   KI5 - 위 사진과 같이 포착 부위를 찾아서 치료한다. - 수천혈과 비슷하다.

7절. 안면부 포착신경 병증

두면부하기 전에 안면부를 했어야 하는데, 깜빡하고 못 했다. 안면부 환자도 많이 오니 잘 공부하자

1. 안와상신경 포착신경병증 (supraorbital nerve entrapment)

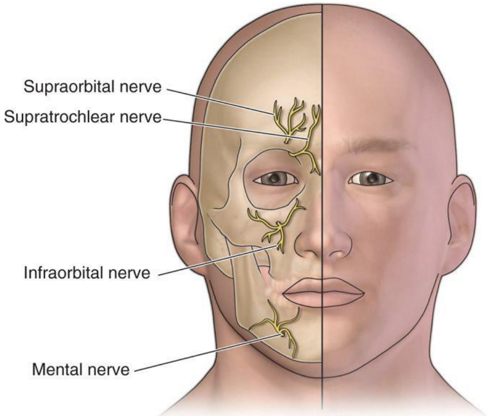
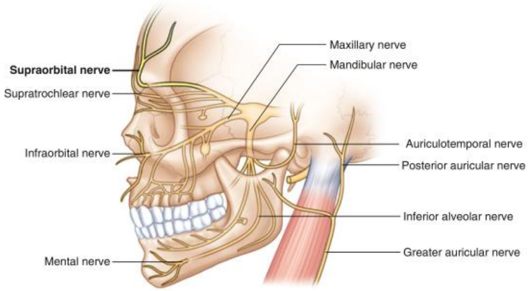
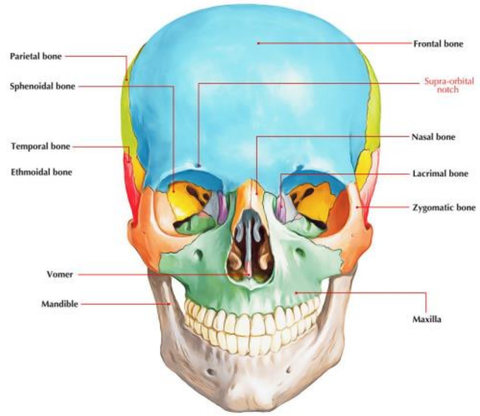
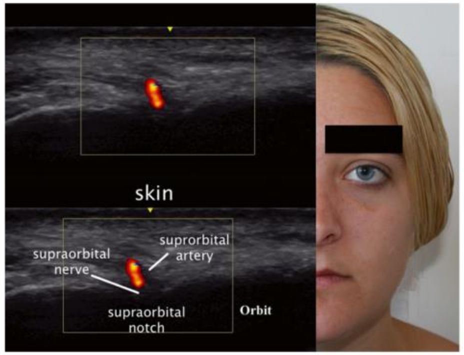
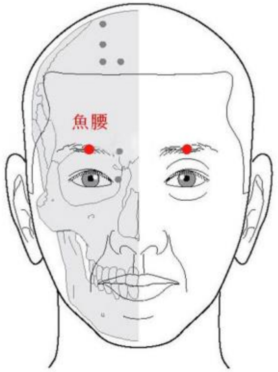
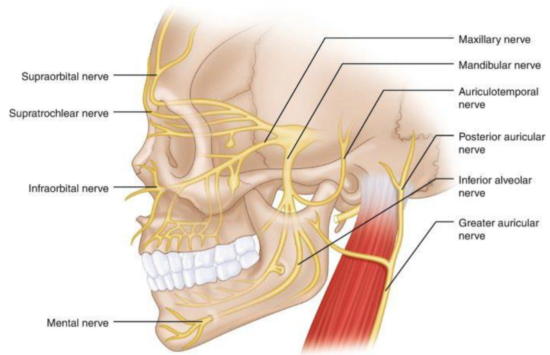
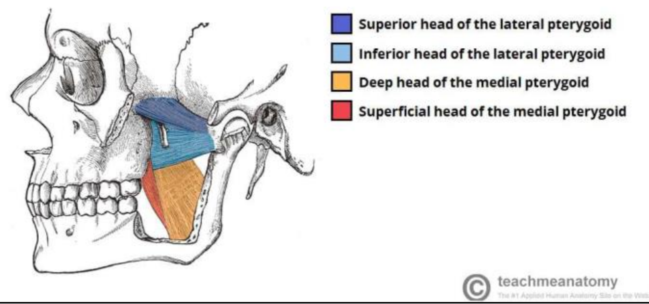
<table>
<tr><td>증상</td><td>•눈위 이마 부위-전두부 통증
•다른 형태의 두통과 같이 있을 수 있음
•시야 흐림, 안구 통증 등의 증상이 동반
•Extracranial headache 원인의 4%

</td></tr>
<tr><td>원인</td><td>•외상
•수경, 헬멧의 착용
- 혈관과 같이 분포한다.

</td></tr>
<tr><td>포착 위치</td><td>•Supraorbital notch

</td></tr>
<tr><td>통증 부위</td><td>- 이마 부위에서 통증을 호소한다.
- 대부분은 긴장성 두통이긴 한다.
- 수경을 자주 쓰는 수영선수에게서 자주 나타난다.

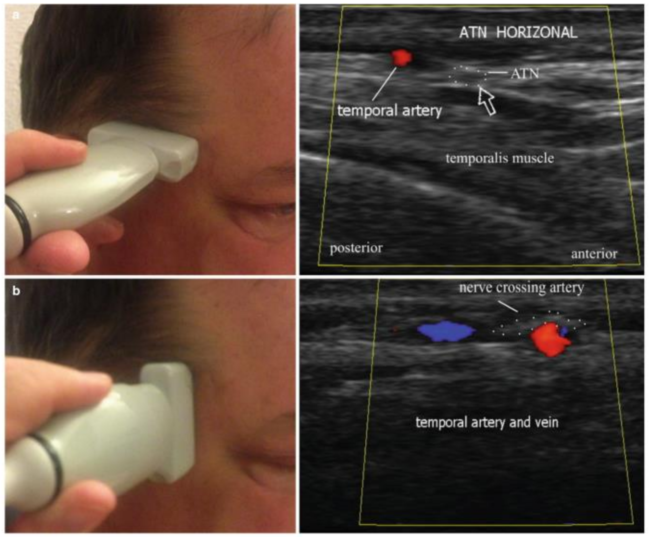
Fig. 14.1 Patient pain complaint from supraorbital nerve entrapment (Image courtesy of Andrea Trescot, MD)

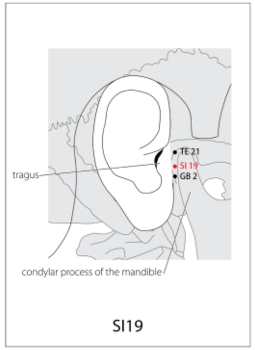
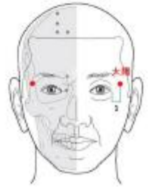
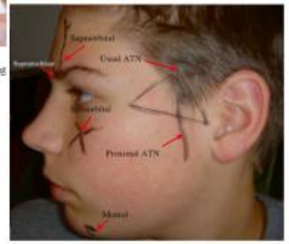
</td></tr>
</table>

영상	- 눈썹 부위에서 접근한다. **Fig. 14.9** Ultrasound image of the supraorbital nerve (Ultrasound image courtesy of David Spinner, MD; composite created by Andrea Trescot, MD) 
추가 내용	 **Fig. 14.7** Supraorbital landmark-guided injection (Image courtesy of Andrea Trescot, MD)  - 시험 공부할 때 스트레스를 많이 받으면 , 눈썹 부위에서 통증이 유발될 수 있다 - 어요혈을 치료포인트로 한다.

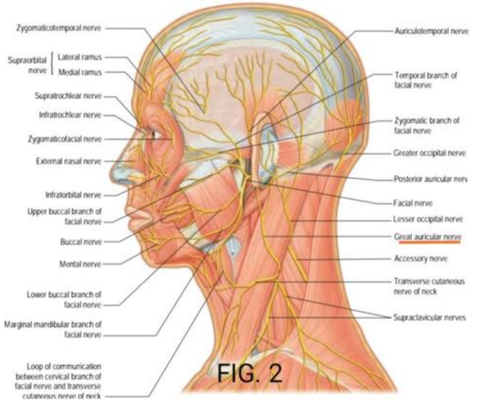
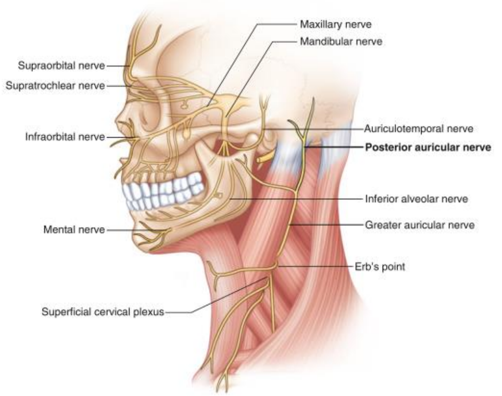
2. 이개측두신경 포착신경병증 (auriculotemporal nerve entrapment)

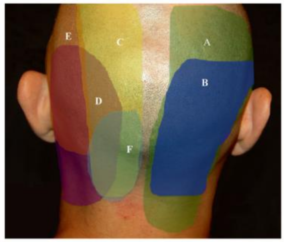
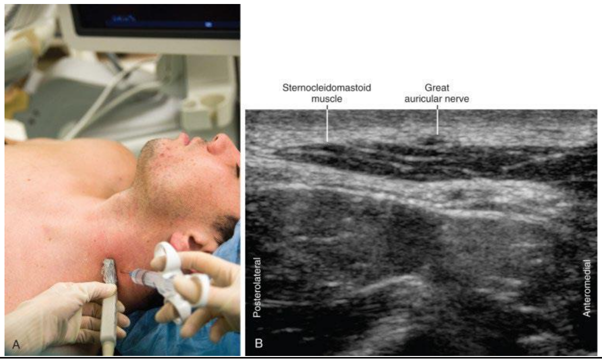
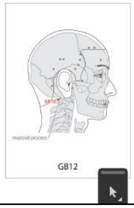
증상	•삼차신경에 의한 두통의 가장 흔한 원인 •이개와 측두 부위의 갑작스러운 통증 발작 (auriculotemporal neuralgia) •음식을 먹거나 음료를 마실 때 한쪽 뺨과 귀에서 과도한 땀과 홍조가 발생 (Frey's syndrome, auriculotemporal syndrome, 드뭄) - 편두통과 비슷하다. 
원인	•TMJ의 병변, 외상, 수술 등 •삼차신경통
포착 위치	Lateral pterygoid muscle 주변 

통증 부위	 Fig. 15.1 Patient's description of pain from auriculotemporal neuralgia (Image courtesy of Andrea Trescot, MD) the preauricular area or by chewing, and can be associated  Fig. 15.2 Frey syndrome. Note the well-healed parotid scar and the facial flushing with eating (Image courtesy of Andrea Trescot, MD) - 측두 쪽으로 가는 신경들이 턱관절 관련 근육들과 연관되어 포착이 나타난다.(두통, 턱 통증 등)
영상	 Fig. 15.10 Ultrasound evaluation of the auriculotemporal nerve. (a) Horizontal evaluation; (b) Vertical evaluation (Images courtesy of Andrea Trescot, MD)

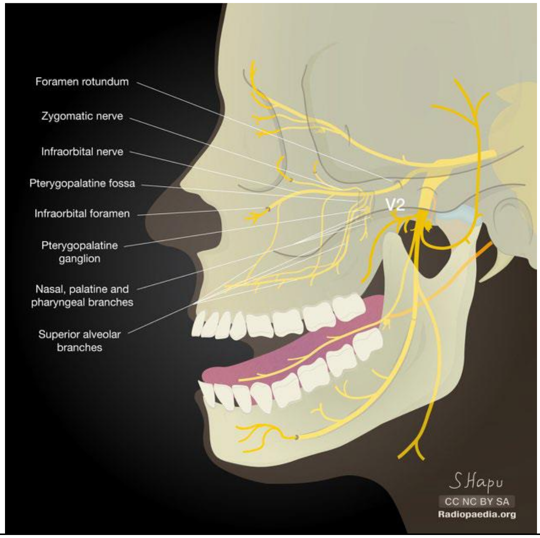
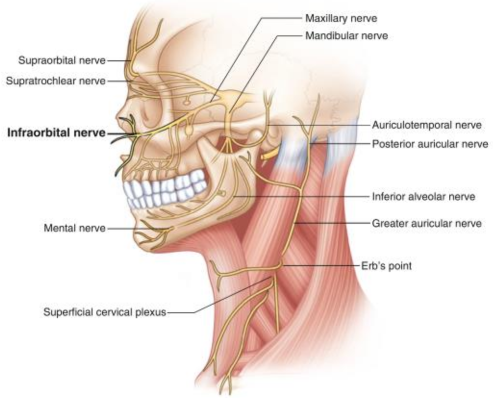
추가 내용	 Fig. 15.8 Auriculotemporal injection (Imag Trescot, MD) • Lan are TM. mai the    Fig. 15.7 Auriculotemporal site of injection (usual and proximal), including other sites of nerve injection (Image courtesy of Andrea Trescot, MD) - 청궁혈, 태양혈을 치료포인트로 한다.

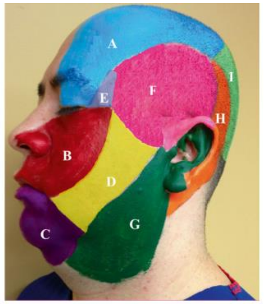
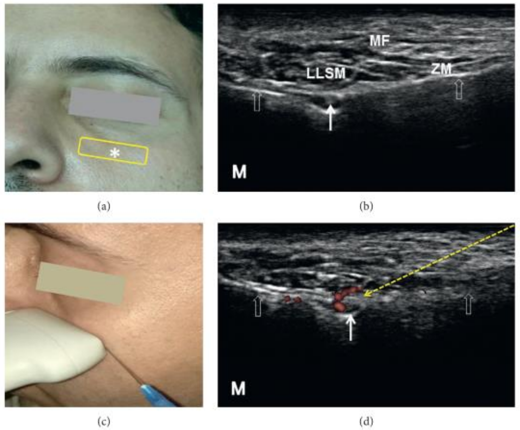
3. 대이개/후이개신경 포착신경병증 (great auricular/posterior auricular nerve entrapment)

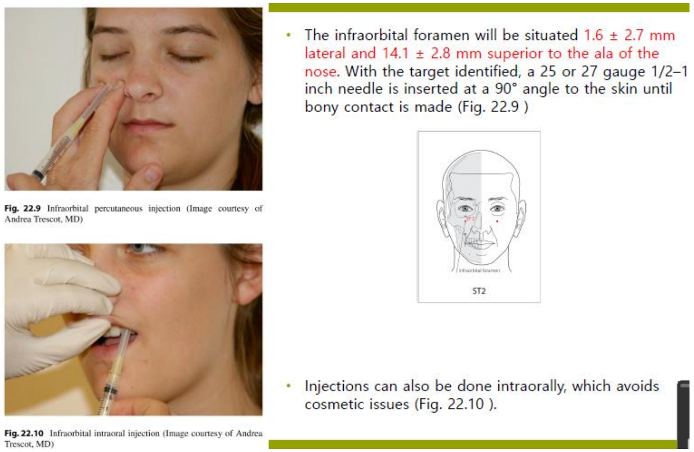
증상	•귀충만감, 청력감소, 이명, 현훈 등 •후두부나 측면의 두통  FIG. 2 - SCM과 관련있다.
원인	•Mastoid 부분의 수술이나 외상 •Mastoid 부분의 압박 **Fig. 16.2** Posterior auricular and great auricular nerve anatomy, lateral view (Image by Springer) 
포착 위치	•SCM의 후면부

통증 부위	 **Fig. 16.1** Pattern of pain from posterior auricular neuralgia/great auricular neuralgia (Image courtesy of Andrea Trescot, MD)  **Fig. 16.8** Pattern of posterior cervical and occipital pain. *A* atlantoaxial joint, *B* atlantooccipital joint, *C* greater occipital nerve (GON), *D* posterior auricular nerve (PAN), *E* lesser occipital nerve (LON), *F* [illegible] occipital nerve (TON) (Image courtesy of Andrea Trescot, MD)
영상	
추가 내용	완골혈을 치료포인트로 한다.  **Fig. 16.9** Landmark-guided posterior auricular/great auricular nerve injection (Image courtesy of Andrea Trescot, MD) • The thumb is then placed along the mastoid of the affected side. With the thumb, feel for the vertical groove between the mastoid process and the occiput.  **Fig. 16.7** Posterior auricular/great auricular nerve examination (Image courtesy of Andrea Trescot, MD) 

4. 안와하신경 포착신경병증 (infraorbital nerve entrapment)

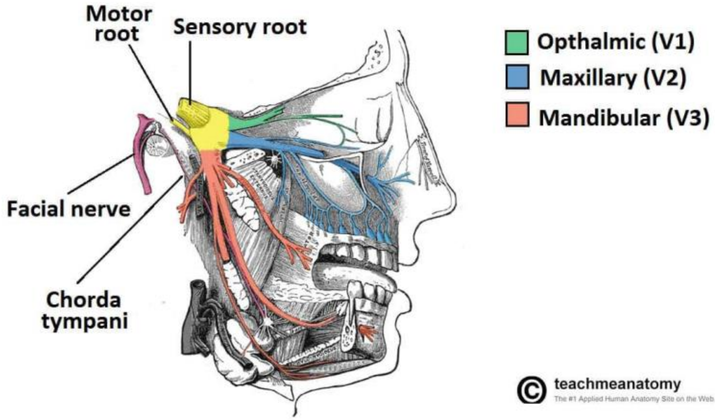
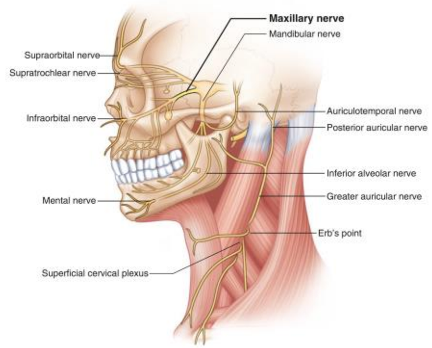
증상	•상악의 통증 •위 치아, 코, 하안검으로 방사되는 통증 
원인	•광대뼈나 치아의 손상 •삼차신경통 - 보통 삼차신경통 진단을 받고 오는 경우가 많다. **Fig. 22.2** Infraorbital anatomy (Image courtesy of Springer) 

포착 위치	•Infraorbital foramen (infraorbital artery에 의한 압력이 증가 될 때)
통증 부위	안면부 포착신경병증 – 안와하신경 포착신경병증  **Fig. 22.1** Pattern of pain from infraorbital neuralgia (Image courtesy of Andrea Trescot, MD)  **Fig. 22.5** Sensory areas of the trigeminal and cervical nerve branches: *A* supraorbital nerve, *B* infraorbital nerve, *C* mental nerve, *D* buccal nerve, *E* lacrimal nerve, *F* auriculotemporal nerve, *G* superficial cervical plexus, *H* posterior auricular nerve/great auricular nerve, *I* occipital nerve (Image courtesy of Terri Dallas-Prunskis, MD)
영상	- 아래 사진과 같이 프로브를 대며, 동맥이 지나가는 것을 확인  FIGURE 4: Sonoanatomy and ultrasound-guided injection technique for the infraorbital nerve: (a) the transducer position (yellow rectangle), (b) ultrasound image of the infraorbital nerve (white solid arrow) from the infraorbital foramen, (c) introducing the needle in the lateral-to-medial direction using the in-plane approach to target the infraorbital nerve, and (d) power Doppler image of the infraorbital vessels. The asterisk (*) denotes the infraorbital foramen on the face. The empty white arrows denote the bony cortex of the maxilla. The yellow dashed arrow denotes the needle trajectory. LLSM: levator labii superioris muscle; ZM: zygomaticus minor muscle; MF: malar fat; M: medial side. All the pictures were obtained from the face of the first author. 한다.

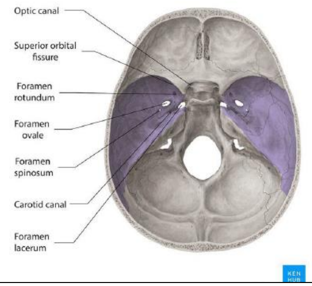
<table>
<tr><td>추가 내용</td><td>- 피부 부위에 수술하기 때문에 트러블이(멍) 생길 수 있다고 고지해야 하며, 아래 사진과 같이 입 안쪽으로 시술하는 경우도 있


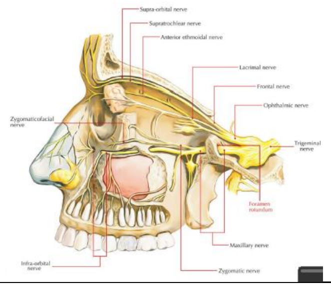
다.</td></tr>
</table>

5. 상악신경 포착신경병증 (maxillary nerve entrapment)

<table>
<tr><td>증상</td><td>•볼(상악) 부위의 통증과 감각이상
•윗 입술이나 입안으로 방사되는 통증
•웃거나 세수하거나 하는 등 안면이 자극되면 증상이 발생
•삼차신경통, 안와하신경통과 유사

</td></tr>
<tr><td>원인</td><td>•광대뼈 혹은 상악의 골절/외상
•기타 압박을 일으킬 수 있는 병변
Fig. 23.2 Anatomy of the trigeminal nerve and cervical nerves (Image courtesy of Springer)

</td></tr>
<tr><td>포착 위치</td><td>•Foramen rotundum</td></tr>
</table>

•Infraorbital foramen





통증 부위

부 포착신경병증 – 상악신경 포착신경병증

Fig. 23.1 Pattern of maxillary nerve pain (Image courtesy of Andrea Trescot, MD)

Fig. 23.3 Distribution of trigeminal nerve sensation. *V1* ophthalmic division/supraorbital nerve, *V2* maxillary division/maxillary nerve, *V3* mandibular division/mandibular nerve (Image courtesy of Terri Dallas-Prunskis, MD)

영상

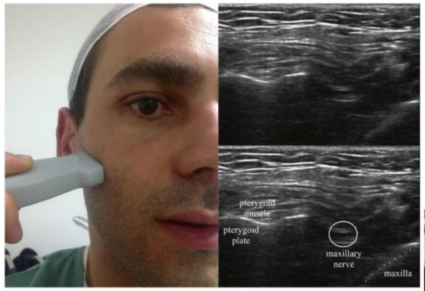


Fig. 23.10 Ultrasound image of the maxillary nerve (Image courtesy of Thiago Nouer, MD)

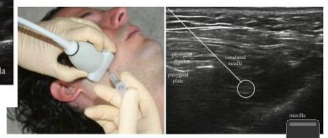


Fig. 23.11 Ultrasound-directed injection of the maxillary nerve (Image courtesy of Andrea Trescot, MD)

추가 내용

- 하관혈을 치료포인트로 잡는다.
- 턱관절 질환에서 함께 치료한다.
- 어금니 안쪽으로 더 들어가서 자입한다.

- The cheek at the angle of the mouth is retracted until the first upper molar is seen. The needle is introduced through the mucosa over the tooth and advanced backward, passing tangential to the maxillary tuberosity. When contact with the bone is lost at a depth of 3–4 cm, the needle is then advanced 0.5 cm more.

Fig. 23.8 Intraoral injection of the maxillary nerve (Image courtesy of Andrea Trescot, MD)

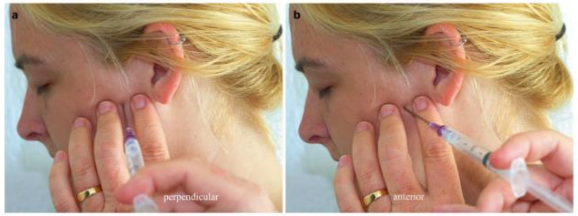
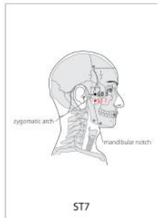
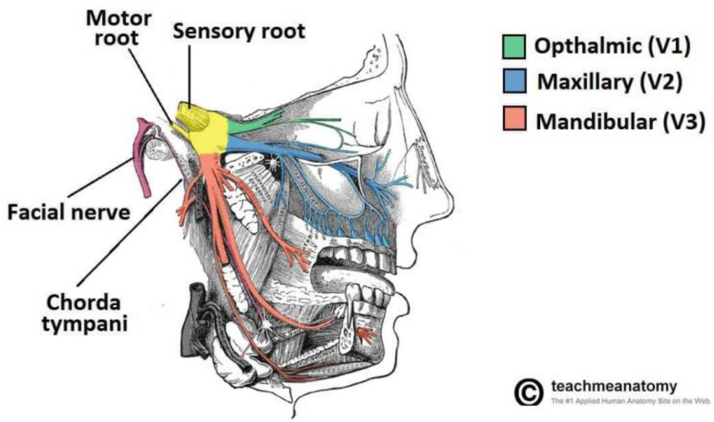
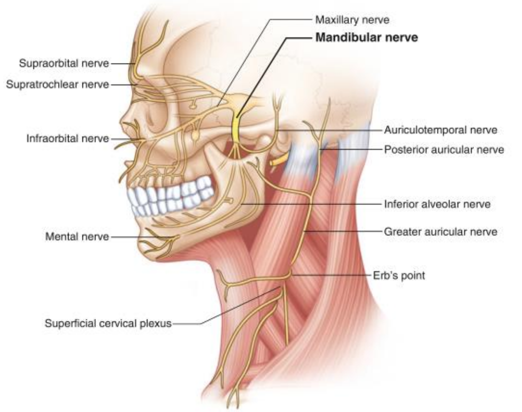


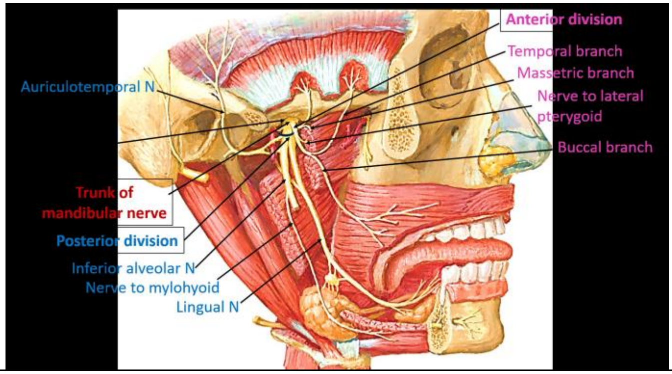
Fig. 23.7 (a) A 22-gauge needle is placed perpendicular to the skin at the posterior and inferior aspects of the coronoid notch. (b) The needle is advanced to the lateral pterygoid plate, then redirected anteriorly and superiorly. Extraoral injection of the maxillary nerve (Image courtesy of Andrea Trescot, MD)



After a skin prep and subcutaneous local anesthetic infiltration, a 25-gauge 2 inch or 22-gauge 3.5 inch needle is placed perpendicular to the skin at the posterior and inferior aspects of the notch (Fig. 23.7a), which should be close to the middle of the zygoma. The needle is advanced until it encounters the lateral pterygoid plate at a depth of 4–5 cm. The needle is then redirected anteriorly and superiorly toward the upper root of the nose (Fig. 23.7b).

6. 하악신경 포착신경병증 (maxillary nerve entrapment)

증상	•아래턱 부위, 아래 치아 부위의 통증 •Auriculotemporal nerve 부위의 통증도 같이 있을 수 있음 •껍을 씹거나 하품할때 통증이 발생 
원인	•광대뼈 혹은 상악/하악의 골절/외상 •기타 압박을 일으킬 수 있는 병변 - 밑에 여러 신경, 혈관들이 지나가기 때문에 문제가 잘 발생할 수 있다. **Fig. 24.4** Nerve anatomy of the face (Image courtesy of Springer) 

포착 위치	•Foramen ovale •Infratemporal fossa 
통증 부위	안면부 포착신경병증 – 하악신경 포착신경병증  **Fig. 24.1** Pattern of mandibular nerve pain (Image courtesy of Andrea Trescot, MD)  **Fig. 24.2** Distribution of trigeminal nerve sensation. *V1* ophthalmic division/supraorbital nerve, *V2* maxillary division/maxillary nerve, *V3* mandibular division/mandibular nerve (Image courtesy of Terri Dallas-Prunskis, MD)

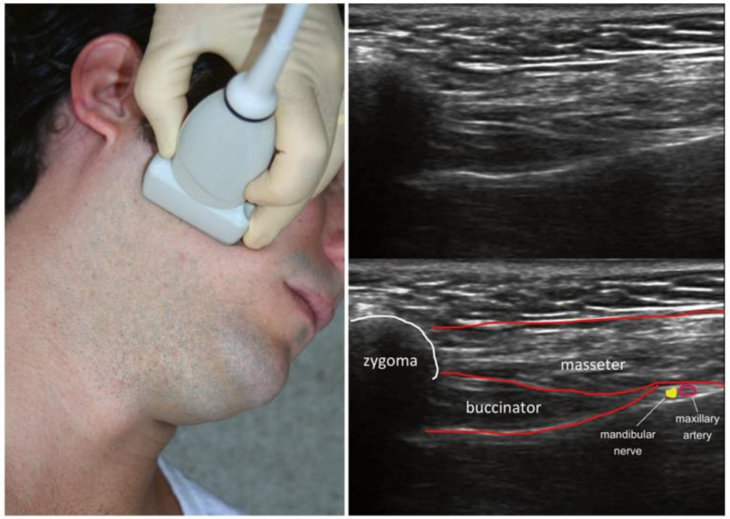
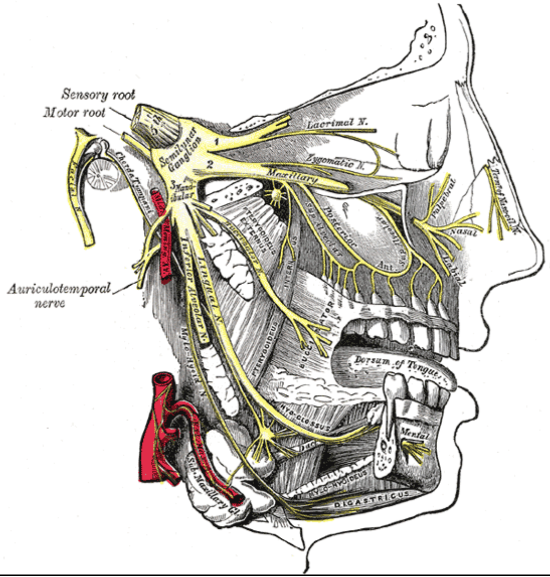
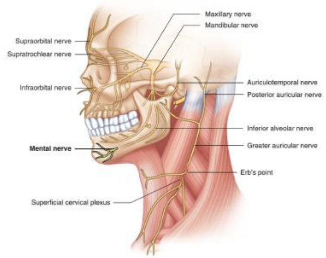
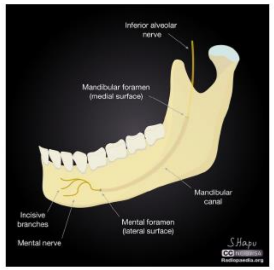
<table>
<tr><td>영상</td><td>

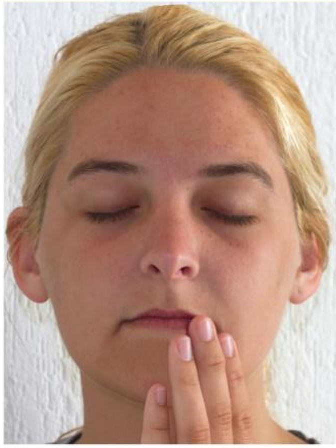
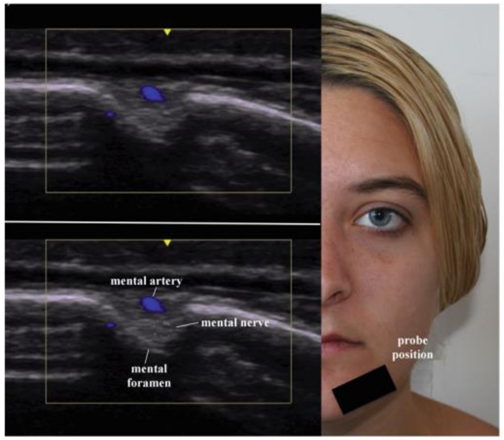
Fig. 24.14 Ultrasound evaluation of the mandibular nerve (Image courtesy of Andrea Trescot, MD)</td></tr>
<tr><td>추가 내용</td><td>- 앞의 신경보다는 좀 더 뒤쪽으로 자입한다.


Fig. 24.9 (a, b) Extraoral injection of the mandibular nerve (Image courtesy of Andrea Trescot, MD)</td></tr>
</table>

7. 턱끝신경 포착신경병증 (mental nerve entrapment)

<table>
<tr><td>증상</td><td>•아래턱 일부, 아래 입술의 감각장애위주, 통증도 있을 수 있음
•치과 치료 시 마취 후 느낌과 비슷
•1830년에 처음 기록되어 있음

</td></tr>
<tr><td>원인</td><td>•Dental/facial trauma
•기타 압박을 유발할 수 있는 병변
Fig. 25.2 Mental nerve anatomy (Image courtesy of Springer)



</td></tr>
<tr><td>포착 위치</td><td>•Mental formane</td></tr>
</table>

통증 부위	 **Fig. 25.1** Pattern of pain from mental neuralgia (Image courtesy of Andrea Trescot, MD)  **Fig. 25.5** Sensory areas of the trigeminal and cervical nerve branches: *A* supraorbital nerve, *B* infraorbital nerve, *C* mental nerve, *D* buccal nerve, *E* lacrimal nerve, *F* auriculotemporal nerve, *G* superficial cervical plexus, *H* posterior auricular nerve/great auricular nerve, *I* occipital nerve (Image courtesy of Terri Dallas-Prunskis, MD)
영상	- 이런 식으로 프로브를 놓는다. **Fig. 25.13** Ultrasound visualization of the mental nerve (Image courtesy of David Spinner, MD) 

추가 내용	- 마찬가지로 입 안쪽으로 시술한다.  **Fig. 25.9** Percutaneous (extraoral) mental nerve injection (Image courtesy of Andrea Trescot, MD)  **Fig. 25.10** Intraoral mental nerve injection (Image courtesy of Andrea Trescot, MD)

나중에 졸업하고 로컬 나가면 해당 내용들을 다시 찾아볼 것이라 본다. 포착 병증이 차지하는 비율이 매우 적긴 하다.

다음 페이지의 논문들은 교수님이 소개하시려고 했다가 다음 시간이 시험이라서 넘어가심

Nerve entrapment syndromes: detection by ultrasound

ULTRA SONO GRAPHY

Christoph Schwabl[1], Gernot Schmidle[2], Peter Kaiser[2], Elena Drakonaki[3], Mihra S. Taljanovic[4], Andrea S. Klauser[1]

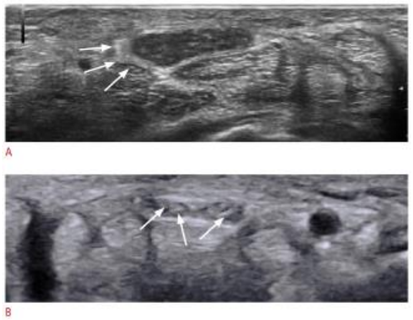


Fig. 1. A 78-year-old patient with unilateral symptoms of carpal tunnel syndrome.
A. The ultrasonography examination reveals a swollen, hypoechoic nerve with a thickened perineurium on the affected side (arrows). B. The other hand shows a normal median nerve inside the carpal tunnel with a typical honeycomb-like appearance, with hypoechoic nerve fascides and hyperechoic interfascicular perineurium (arrows).

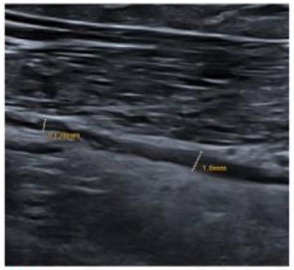


Fig. 6. A 42-year-old patient presenting with pain at the radial aspect of the cubital fossa. Ultrasonography shows an irregular hypoechoic radial nerve within the arcade of Frohse.

포착신경병증

Ultrasound-guided acupotomy release for treating common peroneal nerve entrapment syndrome: a case description

Yu-Yang Ou[1,2], Yun-Nan Li[2], Xiao-Jie Sun[2], Shi-Liang Li[1]

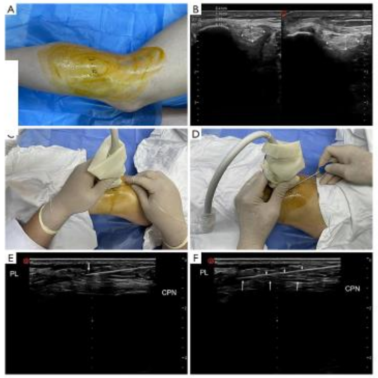


Figure 1 Preoperative and intraoperative photographs and ultrasound images of acupotomy treatment. (A) Preoperative approximate positioning and disinfection. (B) Preoperative comparison of the cross-sectional areas of the bilateral CPN. The left image is the healthy side, and the right image is the affected side. The highly echogenic plane corresponds to the surface of the FH. The honeycomb-like echogenic pattern is the CPN. On the affected side, the short-axis distance was 0.41 cm, and the long axis distance was 1.02 [illegible] healthy side, the short-axis distance was 0.45 cm, and the long-axis distance was 0.72 cm. (C) Local anesthesia in the longitudi[illegible] Acupotomy insertion in the longitudinal plane. (E,F) Longitudinal ultrasonographic image of needle-knife treatment. (E) Loo[illegible] fibular canal entrance and the surrounding soft tissues with a needle knife. The arrow indicates the fibular canal entrance. ([illegible] of the PL with a needle knife reaching the surface of the CPN. The short arrows indicate the needle knife, while long arrows [illegible] surface of the CPN. CPN, common peroneal nerve; FH, fibular head; PL, peroneus longus.

Study Protocol Systematic Review

Medicine

OPEN

Pharmacopuncture for nerve entrapment syndrome
A protocol for systematic review

Jin-Ho Jeong, PhD, KMD[a], Ji Hye Hwang, PhD, KMD[b,*]

Abstract

Background: Nerve entrapment syndrome occurs when the nerves become compressed or entrapped and restricted. This study aims to evaluate the effectiveness and safety of pharmacopuncture in patients with nerve entrapment syndrome.

Methods: A search will be conducted from inception to August 2022 using the following 11 electronic databases: MEDLINE, EMBASE, Cochrane Central Register of Controlled Trials, Allied and Complementary Medicine Database China National Knowledge Infrastructure, and 6 Korean databases. All randomized controlled trials (RCTs) evaluating pharmacopuncture treatment for various nerve entrapment syndromes will be considered, with no restrictions regarding the type of pharmacopuncture solution used. Two reviewers will perform the data extraction and quality assessment using a predefined data extraction form. The methodological quality of the included RCTs will be assessed using the Cochrane risk-of-bias tool.

Results: This systematic review will provide high-quality evidence to determine the efficacy and safety of pharmacopuncture therapy for nerve entrapment syndrome.

Conclusion: Our findings will be informative for patients with nerve entrapment syndrome, as well as clinicians, policymakers, and researchers.

Abbreviations: CI = confidence interval, KM = Korean Medicine, RCT = randomized controlled trial.

Keywords: acupoint injection, entrapment neuropathy, nerve entrapment, pharmacopuncture, protocol, systematic review

Study Protocol Systematic Review

Medicine

OPEN

Acupotomy for nerve entrapment syndrome
A systematic review protocol

Yifeng Shen, PhD[a], Ting Li, MS[b], Tao Cai, MD[c], Juan Zhong, PhD[a], Jing Guo, PhD[a], Huarui Shen, MS[d,*]

Abstract

Background: Acupotomy has been widely used to treat nerve entrapment syndrome. But its efficiency has not been scientifically and methodically evaluated. The aim of this study is to evaluate the efficacy and safety of the acupotomy treatment in patients with nerve entrapment syndrome.

Methods: Fifteen databases will be searched from inception to Dec 2019. We will include randomized controlled trials (RCTs) assessing acupotomy for nerve entrapment syndrome. All RCTs on acupotomy or related interventions will be included. Study inclusion, data extraction and quality assessment will be performed independently by 2 reviewers. Assessment of risk of bias and data synthesis will be performed using RevMan 5.3 software. Cochrane criteria for risk-of-bias will be used to assess the methodological quality of the trials.

Results: This study will provide a high-quality synthesis of pain VAS and functional disability or the quality of life, the success treatment rate, the recurrent rate and the complications rate to assess the effectiveness and safety of acupotomy for nerve entrapment syndrome patients.

Conclusion: This systematic review will provide evidence to judge whether acupotomy is an effective intervention for patients with nerve entrapment syndrome.

PROSPERO registration number: CRD42018109086.

Abbreviations: 95% CI = 95% confidence interval, GRADE = Grading of Recommendations Assessment, Development and Evaluation, ICTRP = WHO International Clinical Trials Registry Platform, MD = mean difference, RCTs = randomized controlled trials, RR = relative risk, VAS = visual analog scale.

Keywords: acupotomy, nerve entrapment syndrome, protocol, systematic review